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Amrita School of Physical Science Science,

Amrita Vishwa Vidyapeetham

Department of Mathematics

Int MSc Degree (Supplementary) Examination, May 2024

S2 Integrated MSc Physics with Minor in Scientific Computing

18MAT 117-Introduction to Ordinary Differential Equations

Time-3 hours

Maximum marks:100

Answer all questions

Qn No.		Marks	CO	Blooms Taxonomy level
1	a) Find a general solution of the ODE $\frac{dy}{dx} = 4\cos 2x$. b) Find the Wronskian and hence verify the functions $y_1 = x^2, x^2 \ln x$ are linearly independent or not	5+5	CO1	L1
2	a) Determine a solution of the Initial value problem $\frac{dy}{dx} + y = \sin 2x,\ y(1) = -1$ b) Show that $\cos 2x$ is an integrating factor of $2\cos y dx = \tan 2x \sin y dy$.	5+5	CO1	L1
3	a) Find a general solution of $y'' + 5y' + 6y = 2e^{-x}$ using method of undetermined coefficient b) Verify $(2xy + 6x)dx + (x^2 + 4y^3)dy$ is exact or not and if yes, find the general solution.	5+5	CO2	L2
4	a) Solve $y'' + 4y = 8x^2$ using method of variation of parameters b) Determine a general solution of Euler-Cauchy equation $x^2y'' - 3xy' + 10y = 0$.	5+5	CO2	L2
5	a) Determine the orthogonal trajectories of the family of curves $y = cx$, where c is a constant	5+5	CO2	L2

	b) Find a basis of solutions of the differential equation $(x^2 - x)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = 0$, if one solution is $y_1 = x$.			
6	a) Find a second order homogeneous linear Ordinary differential equation for which the functions $y_1 = 1,\ y_2 = e^{3x},\ y(0) = 2,\ y'(0) = -1$ are the solutions b) Determine the type and stability of the critical point. Then find a general solution of the system of equations $y^1{}_1 = -y_1 - y_2,\ y^1{}_2 = -4y_1 - y_2$.	5+5	CO3	L2
7	a) Express the function $f(t)$ in terms of unit step function $f(t) = t,\ t > 1$ b) Determine the inverse Laplace transform of (i) $L^{-1}\left\{\frac{1}{s(s+1)}\right\}$ (ii) $L^{-1}\left\{\frac{3s}{s^2-4}\right\}$	5+5	CO4	L1
8	a) Using Laplace transform solve the integral equation $y(t) - \int_0^t y(\tau)\sin(t-\tau)\,d\tau = 1$. b) Using Laplace transform method determine a general solution of the differential equation $\frac{dy}{dt} + 2y = 0,\ y(0) = 1.5$.	5+5	CO4	L1
9	Using Laplace transform method determine a general solution of the differential equation $\frac{d^2y}{dt^2} - 3\frac{dy}{dt} + 2y = 4t - 8,\ y(0) = 2,\ y'(0) = 7$	10	CO4	L2
10	Find the general solution of the system $\begin{cases} \frac{dy_1}{dt} = y_2 \\ \frac{dy_2}{dt} = -4y_1 \end{cases}$ using Eigen values and Eigen vectors.	10	CO3	L2

L_1-Remember, L_2-Understand, L_3-Apply, L_4-Analyze, L_5-Evaluate, L_6-Create